MIL-D-51256B
8 November 1984
SUPERSEDING
MIL-D-51256A
29 December 1967

# MILITARY SPECIFICATION

## DETECTOR KIT, CHEMICAL AGENT, ABC-M18A2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers one type of chemical agent detector kit.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| NN-P-71 | - | Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial) |
| PPP-B-585 | - | Boxes, Wood, Wirebound |
| PPP-B-621 | - | Boxes, Wood, Nailed and Lock-Corner |
| PPP-F-320 | - | Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes |

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament, Munitions and Chemical Command, ATTN: AMSMC-TDC-S (A), Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6665

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:54Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH AND DEVELOPMENT CENTER

TDPL 5-77-2092 - Detector Kit, Chemical Agent, ABC-M18A2

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

ASTM STANDARDS

D 1193 - Reagent Water
D 3951 - Commercial Packaging

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:54Z
Check the source to verify that this is the current version before use.

## 3. REQUIREMENTS

### 3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL 5-77-2092 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the detector kit shall conform to the specifications and drawings listed on TDPL 5-77-2092 and subsidiary drawings.

3.1.2.1 Sodium hydroxide bottle (Drawing 5-77-2100-1). The sodium hydroxide bottle shall contain 0.68 ± 0.04 gram (g) of ACS grade sodium hydroxide when tested as specified in 4.4.4.1.

3.1.2.2 Tris buffer bottle (Drawing 5-77-2100-2). The tris buffer bottle shall contain at least 8 milliliters (mL) of 0.05 molar tris (hydroxymethyl) aminomethane solution with a pH of 8.0 ± 0.1 when tested as specified in 4.4.4.2.

3.2 Assembly. The kit shall be assembled in accordance with Drawing 5-77-2092.

3.2.1 Discard date of assembled kit. The discard date of an assembled kit shall be 3 years after the earliest manufacturing date of the detector tubes of the kit. The assembled kit shall be submitted to the Government at least 2 years and 8 months before the discard date of the kit.

3.3 Preproduction. Prior to the start of regular production, a preproduction sample of detector kits shall be produced in accordance with this specification for examination and test (see 4.3).

3.3.1 Drop stability.

3.3.1.1 Case. The preproduction chemical agent detector kit cases shall not crack or break when tested as specified in 4.4.4.3.1.

3.3.1.2 Contents. The preproduction chemical agent detector kit components shall not crack or break when tested as specified in 4.4.4.3.2. A maximum of five broken detector tubes will be allowed for each kit.

3.4 Workmanship. The chemical agent detector kit and all components and subassemblies thereof shall be free from defects and from contamination by foreign matter such as dirt, grease, and oil.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:54Z
Check the source to verify that this is the current version before use.

in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3), and
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sampling. A preproduction sample of eight detector kits shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination. Each preproduction kit in the sample shall be examined for the applicable defects listed in 4.4.3.3. The presence of any defect in any sample kit shall be cause for rejection of the preproduction lot.

4.3.2.2 For test. The first test conducted shall be 4.3.3.1, using three kits from the sample. Using the remaining five kits from the sample, conduct the tests specified in 4.3.3.2, 4.4.4.1, and 4.4.4.2, in that order.

4.3.3 Drop stability tests. Drop stability tests shall be conducted as follows:

4.3.3.1 Case. Bring each sample detector kit to a temperature of -40° ± 5°F and drop the kit twice free-fall, from a height of 6.0 (+0.5 or -0) feet onto a 1/2-inch thick plywood sheet on a concrete floor. After the drops, examine the case for evidence of cracking or breaking.

4.3.3.2 Components. Sample kits packed in accordance with Drawing 5-77-2092 shall be dropped onto a 1/2-inch thick plywood sheet resting on concrete. Each kit shall be dropped three times in a random orientation from a height of 5 feet. The components of the kit shall be examined for any signs of breakage. Special attention shall be given to breakage of the detector tubes. A maximum of 5 broken detector tubes shall be allowed in each kit as a result of the drop test.

4.3.4 Acceptance/rejection criteria. Failure of case, components or any of the bottles to meet the requirements of this specification when tested in accordance with 4.3.3 and 4.4.4 shall be cause for rejection of the preproduction lot of detector kits.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:54Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the detector kits produced by one manufacturer at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of chemical components shall be represented in any one lot of kits.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For test. Five bottles containing sodium hydroxide and five bottles containing tris buffer solution shall be randomly selected from each lot of detector kits.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample detector kits shall be examined in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For test. Sample bottles shall be tested as specified in 4.4.4.1 or 4.4.4.2, as applicable. Failure of any bottle to pass the tests specified in 4.4.4 shall be cause for rejection of the lot of detector kits represented.

4.4.3.3 Classification of defects.

(a) Detector Kit, Chemical Agent, ABC-M18A2 (Drawing 5-77-2092).

| Categories | | Defects |
|---|---|---|
| Critical: | | None defined |
| Major: | AQL 1.0 percent defective | |
| 101 | | Component missing or incorrectly assembled |
| 102 | | Interior or exterior marking missing, incorrect, or illegible |
| 103 | | Rubber tubing incorrectly assembled in ticket adapter (Dwg 5-77-792) |
| 104 | | Discard date of kit and components not as specified (3.2.1) |
| Minor: | AQL 2.5 percent defective | |
| 201 | | Component misplaced inside kit |
| 202 | | Hardware coating missing |
| 203 | | Workmanship (see 3.5) |



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:54Z
Check the source to verify that this is the current version before use.

(b) Packaging (see section 5).

| Categories | Defects |
|---|---|
| Critical: | None defined |
| Major: | AQL 4.0 percent defective |
| 101 | Contents per container not as specified |
| 102 | Container not as specified |
| 103 | Container closure not as specified |
| 104 | Fiberboard pads missing or not as specified (when required) |
| 105 | Container damaged |
| 106 | Marking incorrect, missing or illegible |
| 107 | Unitization not as specified |

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Weight of sodium hydroxide in bottle (Dwg 5-77-2100-1). Determine the weight of sodium hydroxide in each bottle in the sample by titrating with a standard acid solution.

4.4.4.2 pH of tris buffer solution (Dwg 5-77-2100-2). Determine the volume and the pH of the tris buffer solution in each bottle in the sample using a graduated glass cylinder and a suitable pH meter, accurate to 0.1 pH units.

5. PACKAGING

5.1 Unit packing, level A. Each detector kit shall be unit packed as shown on Drawing 5-77-2389.

5.2 Packing. Detector kits shall be packed level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Twenty-four detector kits, unit packed as specified in 5.1, shall be packed in a box having inside dimensions of 20.25 inches by 19.0 inches by 18.5 inches. The box shall conform to class 3, style optional of PPP-B-585 with wood preserved and dried as specified for wood of a grade A box of PPP-B-621. The boxes shall be designed for a type 2 load. All inside faces of the box shall be lined with fiberboard pads and additional pads shall be added as needed to prevent motion of the contents. Pads shall be formed from the material conforming to grade V3c of PPP-F-320. Each box shall be closed as specified in the appendix to PPP-B-585.

5.2.2 Level B. Twenty-four detector kits, unit packed as specified in 5.1, shall be packed as shown on Drawing 5-77-2389.

5.2.3 Commercial. Detector kits, unit packed as specified in 5.1, shall be packed commercially in accordance with ASTM D 3951.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:54Z
Check the source to verify that this is the current version before use.

5.3 Palletization. Eight packs of detector kits shall be palletized in accordance with MIL-STD-147, using palletization pattern number 6 as shown therein. The pallet shall conform to NN-P-71, type IV.

5.4 Marking. Level A and B containers shall be marked as shown on Drawing 5-77-2390. Any additional marking shall be in accordance with MIL-STD-129. Commercial packing containers shall be marked in accordance with ASTM D 3951.

6. NOTES

6.1 Intended use. This kit is intended for use in detecting chemical agents in field concentrations.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification,
(b) Level of packing required (see 5.2), and
(c) Preproduction:

(1) Time allowed for contractor's submission of samples for Government test and evaluation after award of contract,
(2) Name and address of test facility and shipping instructions when testing is performed by the Government, and
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 Related documents. The following documents support the ABC-M18A2 detector kit:

| | |
|---|---|
| Packet, Sodium Pyrophosphate Peroxide | MIL-P-50014 |
| Tablets, Compound 34 | MIL-T-51023 |
| Detector Tubes, Chemical Agent | MIL-D-51082 |
| Detector Ticket, Anticholinesterase Agent | MIL-D-51083 |
| Substrate Solution and Dispenser | MIL-S-51189 |
| Paper, Chemical Agent Detector, VGH, ABC-M8; Booklet of | MIL-P-51409 |

Custodians:

Army - EA
Navy - SH
Air Force - 99

Preparing activity:

Army - EA

Project No. 6665-0469

Review activity:

Air Force - 82

User activity:

Navy - OS



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:54Z
Check the source to verify that this is the current version before use.

**INSTRUCTIONS:** In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

*NOTE* This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

*(Fold along this line)*

*(Fold along this line)*

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

**OFFICIAL BUSINESS**
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12062 WASHINGTON D C
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armament, Munitions and Chemical Command
ATTN: AMSMC-TDS-S (A)
Aberdeen Proving Ground, MD 21010-5423

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:54Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions -- Reverse Side)*

| 1. DOCUMENT NUMBER | 2 DOCUMENT TITLE |
|---|---|
| MIL-D-51256B | DETECTOR KIT, CHEMICAL AGENT, ABC-M18A2 |

| 3a NAME OF SUBMITTING ORGANIZATION | 4 TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| | ☐ USER |
| b ADDRESS *(Street, City, State, ZIP Code)* | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)* |

5 PROBLEM AREAS

a. Paragraph Number and Wording.

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8 DATE OF SUBMISSION *(YYMMDD)* |

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:54Z
Check the source to verify that this is the current version before use.